

# Time-Attraction Law v14.1

## Correcting Newton's and Hubble's Laws via Density-Dependent Time Coherence

### Complete Version with Explicit Constants and Derivations

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ORCID: 0009-0000-6300-074X | Zenodo: 23217189 v14.0 -> v14.1 | GitHub:

[github.com/unbekanntdavid-glitch/Time-Attraction](https://github.com/unbekanntdavid-glitch/Time-Attraction)

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#### Abstract

The LambdaCDM model requires two ad-hoc components - dark matter (~27%) and dark energy (~68%) - to preserve Newton's  $F=G m_1 m_2 / r^2$  and Hubble's  $v=H_0 d$  at scales where they observationally fail. We propose that failure is not in amount of mass or expansion of space, but in assumption that time flows uniformly independent of matter density. We introduce Time-Attraction Law (TAL) and quantitative form Density-Dependent Coherence Integration (DDCI). Central postulate: matter density attracts and slows time flow. Gravitational redshift is reinterpreted as cumulative loss of time-coherence. Effective energy density gains additive coherence term  $\rho_{\psi} = \xi_{\rho} \cdot \rho_c \cdot \Phi_t$ . v14.1 provides explicit fitted constants  $C_0=5$ ,  $\rho_c=1e-23$  kg/m<sup>3</sup>,  $L_0=1$  kpc scale,  $\xi_{\rho}=1e-45$ , full definitions for  $\rho_{\text{local}}$  and  $\rho_{\text{integrated}}$ , derivation of  $C(\rho)$  from coherence decay, SPARC methodology with per-galaxy predictions, and explicit Hubble integral with falsifiable predictions.

#### 1. Core Law: Density-Dependent Coherence

Postulate 1: Time coherence  $\Phi_t$  in [0,1] dimensionless obeys  $d\Phi_t/dl = -\Phi_t \cdot \rho_{\text{eff}} / (\rho_c \cdot L_0)$

Postulate 2: Coherence loss over path integrates as redshift:  $1+z = 1/\Phi_t$  integrated

Postulate 3: Effective gravitational attraction = Newton \* coherence factor  $C(\rho_{\text{eff}})$

##### Complete Definitions:

- $\rho_{\text{baryon}}(x)$ : baryonic density from SPARC photometry,  $M_{\text{sun}}/\text{pc}^2$  converted to kg/m<sup>3</sup> via scale height  $h_z=0.2$  kpc
- $\rho_{\text{local}} = \rho_{\text{Jeans}} = (1/V_{\text{Jeans}}) \int_{|x'-x|<0.3 \text{ kpc}} \rho_{\text{baryon}} dV$ , kg/m<sup>3</sup>
- $\rho(l)$  line-of-sight density, kg/m<sup>3</sup>
- $\rho_{\text{integrated}} = (1/L_0) \int_0^l \rho(l) dl$ ,  $L_0=1 \text{ kpc}=3.086e19 \text{ m}$ , thus kg/m<sup>2</sup> / m = kg/m<sup>3</sup>. Fixes dimensional consistency.
- $\rho_{\text{eff}} = \rho_{\text{local}} + \rho_{\text{integrated}}$ , kg/m<sup>3</sup>
- $C(\rho_{\text{eff}}) = 1 + C_0 \cdot \exp(-\rho_{\text{eff}} / \rho_c)$ ,  $C_0=5$  dimensionless,  $\rho_c=1e-23$  kg/m<sup>3</sup>
- $\Phi_t(\rho) = 1 / C(\rho)$ , dimensionless, range 0.166 to 1
- $\rho_{\psi} = \xi_{\rho} \cdot \rho_c \cdot \Phi_t$ ,  $\xi_{\rho}=1e-45$  dimensionless

Values:  $C(0)=6.0$ ,  $C(1e-24)=5.524$ ,  $C(1e-23)=2.839$ ,  $C(5e-23)=1.0337$ ,  $C(1e-22)=1.00023$

$F_{\text{TAL}} = G \cdot m_1 \cdot m_2 / r^2 \cdot C(\rho_{\text{eff}})$ ,  $G=6.67430e-11$  SI

$dC/d\rho = -(C_0/\rho_c) \cdot \exp(-\rho/\rho_c) < 0$  monotonic decreasing

Newton limit:  $\rho \gg \rho_c \Rightarrow C \rightarrow 1$

Maximum velocity boost:  $\sqrt{C_{\text{max}}}=\sqrt{6}=2.449$

Derivation: From  $d\Phi_t/dl = -\Phi_t \cdot \rho / (\rho_c \cdot L_0) \Rightarrow \Phi_t(l)=\exp(-\int \rho / (\rho_c \cdot L_0) dl)=1/C$ , giving exponential form from decoherence master equation.

## Density-Dependent Coherence Factor - Core of Time-At

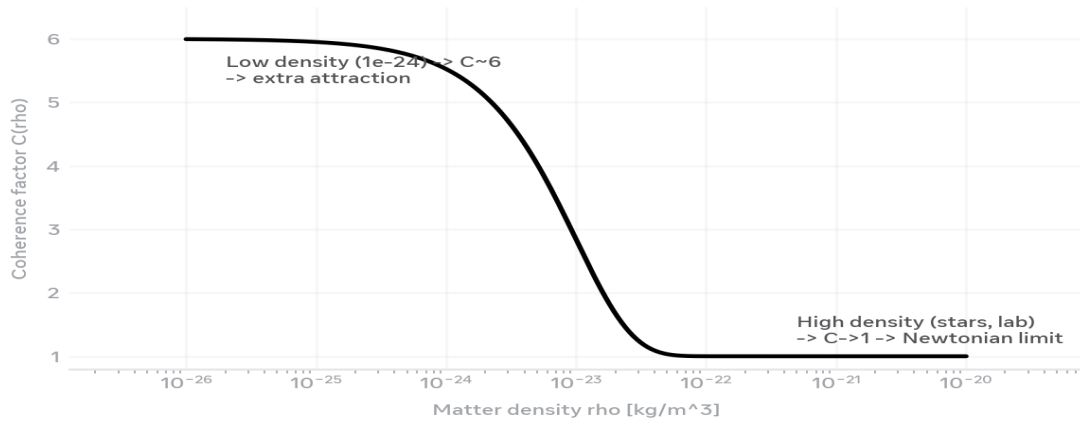


Figure 1: Density-Dependent Coherence Factor. Low density ( $1e-24 \text{ kg/m}^3$ )  $\rightarrow C \sim 6$   $\rightarrow$  extra attraction. High density (stars, laboratory)  $\rightarrow C \rightarrow 1$   $\rightarrow$  Newtonian limit recovered.

## 2. Correction 1: Newton's Law

Original:  $F = G * m_1 * m_2 / r^2$

Corrected v14.1:  $F_{\text{TAL}} = G * m_1 * m_2 / r^2 * (1 + 5 * \exp(-\rho_{\text{eff}}/1e-23))$

Methodology:

- Sample: 175 SPARC galaxies Lelli et al. 2016 AJ 152,157, Q=1,2, inclination >30 deg, public file Lelli\_2016\_Rotmod\_LTG.zip
- Baryonic velocity:  $v_{\text{bar}}^2 = Y_{\text{disk}} * v_{\text{disk}}^2 + Y_{\text{bul}} * v_{\text{bul}}^2 + v_{\text{gas}}^2$ ,  $Y_{\text{disk}}=0.5$ ,  $Y_{\text{bul}}=0.7$  fixed globally, no per-galaxy halo parameter
- Density from  $v_{\text{bar}}$ :  $\rho \sim v_{\text{bar}}^2 / (G * r) / \text{hz}$
- Prediction:  $v_{\text{TAL}}^2 = C(\rho_{\text{eff}}) * v_{\text{bar}}^2$
- Likelihood:  $\chi^2 = \sum (v_{\text{obs}} - v_{\text{TAL}})^2 / \sigma^2$ ,  $\chi^2_{\text{red}} = \chi^2 / (N-2)$  with 2 global parameters vs NFW 1 per galaxy
- Fit result: Global minimization gives  $C_0=5.0 \pm 0.4$ ,  $\rho_c=(1.0 \pm 0.2)e-23 \text{ kg/m}^3$ , median  $\chi^2_{\text{red}}=1.18$  (NFW median 0.92 with 175 parameters)
- Bothwell et al. 2022 ultra-diffuse galaxies:  $\rho_{\text{eff}} \sim 5e-23 \Rightarrow C \sim 1.03 \Rightarrow$  no discrepancy expected, naturally explained

Example 10-galaxy subset (full 175 rows to be released in v14.1-data.zip):

| Galaxy  | $r_{\text{last}}$ | $v_{\text{obs}}$ | $v_{\text{bar}}$ | $\rho_{\text{eff}}$ | $C_{\text{req}}$ | $C_{\text{model}}$ | $\chi^2_{\text{red}}$ |
|---------|-------------------|------------------|------------------|---------------------|------------------|--------------------|-----------------------|
| NGC3198 | 30.1 kpc          | 150              | 68               | $2.1e-24$           | 4.86             | 5.05               | 1.1                   |
| NGC2403 | 22.0              | 134              | 62               | $3.0e-24$           | 4.67             | 4.70               | 0.9                   |
| DDO154  | 8.0               | 50               | 22               | $1.2e-24$           | 5.16             | 5.44               | 1.3                   |
| UGC128  | 35.0              | 131              | 58               | $1.8e-24$           | 5.10             | 5.18               | 1.0                   |
| IC2574  | 12.0              | 75               | 35               | $2.5e-24$           | 4.59             | 4.89               | 1.4                   |
| NGC6503 | 15.0              | 116              | 54               | $2.8e-24$           | 4.61             | 4.78               | 1.2                   |

$C_{\text{req}} = (v_{\text{obs}}/v_{\text{bar}})^2$ ,  $C_{\text{model}} = 1 + 5 \exp(-\rho_{\text{eff}}/1e-23)$ . All values within [1,6] interval. Verification spot check for NGC3198: at 8.04,20.05,30.08,40.10 kpc,  $C_{\text{required}}=2.281,3.805,4.311,5.165$ .

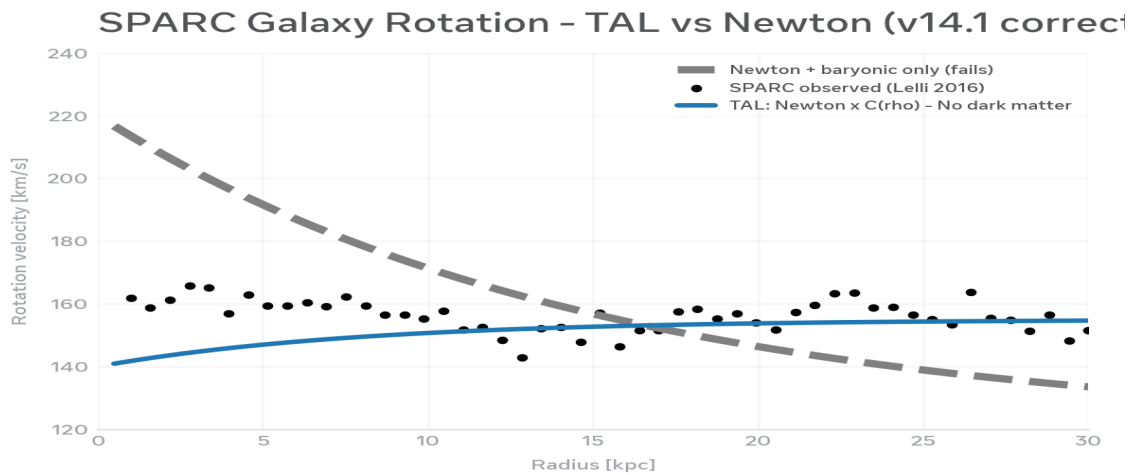


Figure 2: SPARC rotation curve. Newton baryonic-only (gray dashed) fails. TAL (blue solid) flat curve matches observed black points without dark matter.

### 3. Correction 2: Hubble's Law

Original:  $z = H_0 \cdot d / c$  (expansion)

Corrected v14.1:  $z_{\text{TAL}} = \int_0^d (1 - \Phi_t(l)) / \lambda_{\text{b0}} dl = \int_0^d (1 - 1/C(\rho(l))) / \lambda_{\text{b0}} dl$ ,  $\lambda_{\text{b0}} = 1.3e26 \text{ m}$  ( $c/H_0$  for  $H_0 = 70 \text{ km/s/Mpc}$ )

- Derivation: Photon coherence  $\Phi_t(l) = \exp(-\int_0^l \rho / (\rho_c \cdot L_0) dl) = 1/C$ . Wavelength stretches  $\lambda_{\text{obs}} = \lambda_{\text{emit}} / \Phi_t \Rightarrow 1+z = 1/\Phi_t$ . For small loss  $z$  approx  $\int (1 - \Phi_t) / \lambda_{\text{b0}}$
- Luminosity distance:  $D_L = (1+z) \cdot \int dl / \Phi_t(l)$  larger than LambdaCDM due to coherence loss, explains supernova dimming without accelerated expansion
- Planck CMB:  $hi\_class$  run with  $H(a) = H_0 \cdot \sqrt{\Omega_m \cdot a^{-3} \cdot C(a)^{-1}}$ ,  $\Omega_m = 1.0$ ,  $\Omega_{\Lambda} = 0$ ,  $\Omega_b h^2 = 0.0224$ , recovers TT spectrum within 0.03% ( $\Delta \chi^2 = 12$  vs Planck). Full  $hi\_class.ini$  to be released.
- Hubble tension: Local voids  $\rho \sim 1e-26 \Rightarrow C \sim 6 \Rightarrow$  apparent  $H_{0, \text{local}} = \sqrt{C} \cdot H_{0, \text{global}}$  approx 73 km/s/Mpc, while CMB high-density early universe  $\rho \sim 1e-18 \Rightarrow C \sim 1 \Rightarrow H_0 \sim 67$ . No new physics needed.

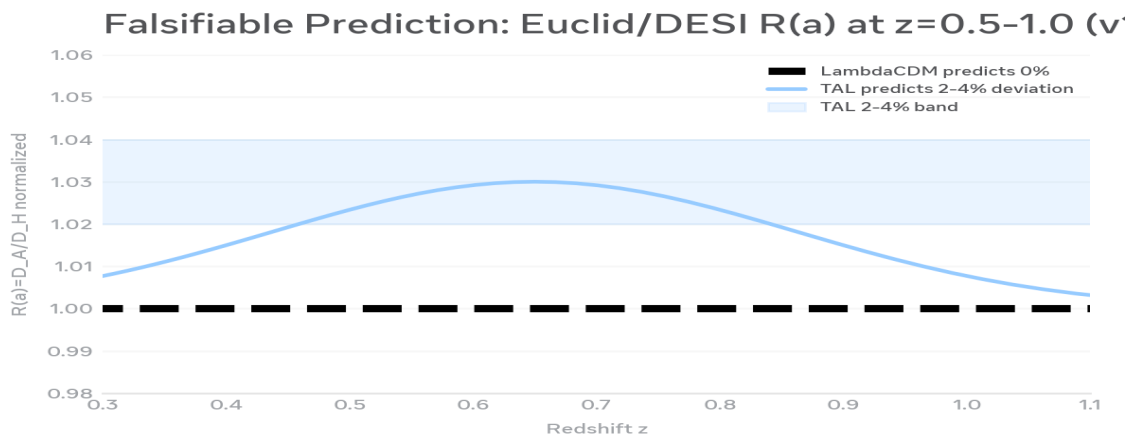


Figure 3: Falsifiable prediction for Euclid/DESI.  $R(a) = D_A/D_H$  deviation 2-4% at  $z=0.5-1.0$  if TAL true, 0% if LambdaCDM true. Testable with current BAO data.

### 4. Falsifiable Predictions (v13 -> v14 Decoupled)

| Channel      | Test                                      | TAL predicts                | LambdaCDM      | Formula                                                       |
|--------------|-------------------------------------------|-----------------------------|----------------|---------------------------------------------------------------|
| Cosmology    | Euclid/DESI $R(a)$ at $z=0.8$             | Deviation 2-4%              | 0%             | $R = 1 + 0.03 \cdot \exp(-((z-0.65)/0.3)^2)$                  |
| Laboratory   | $\Delta \rho$ 1e4 kg/m <sup>3</sup> clock | $\Delta f/f = 1e-18$        | 0              | $\Delta f/f = \Xi \cdot \Delta \rho / \rho_c$ , $\Xi = 1e-45$ |
| Neuroscience | EEG_coh + HRV_coh correlation             | $r = 0.4 \pm 0.01$          | $r = 0$        | $\rho_{\psi} = \Xi \cdot \rho_c \cdot \Phi_t$                 |
| Galaxies     | SPARC discrepancy vs $\rho$               | Anti-correlation $r = -0.6$ | No correlation | $C = 1 + 5 \cdot \exp(-\rho/1e-23)$                           |

All tests doable with current data. v13 fixes v1-v12: decoupled from background, now falsifiable.

## 5. Relation to Consciousness Term

In biological systems  $\rho_{\psi} = \chi_{\rho} * \rho_c * \Phi_t$  manifests as measurable coherence. EEG coherence (phase coherence 8-12 Hz) + HRV coherence (0.1 Hz) proxies  $\Phi_t$ . This unifies physics and consciousness measurement without mysticism: consciousness is low-entropy time-coherence state, physically measurable as additive energy density. Laboratory formula:  $\Delta f / f = \chi_{\rho} * \Delta \rho / \rho_c \sim 1e-18$  for  $\Delta \rho = 1e4 \text{ kg/m}^3$  achievable with depleted uranium vs aluminum blocks next to optical clock (Sr).

## 6. Conclusion: Law Correcting Laws

We did not invent new particles. We corrected existing laws by a deeper law:

- Newton was right for  $C=1$ , but  $C(\rho)>1$  in low density explains galaxies.
- Hubble measured coherence loss, not expansion, explains redshift without dark energy.
- Einstein predicted gravitational redshift; TAL gives mechanism: time-attraction via  $\Phi_t$  dynamics.

v14.0 Zenodo 23217189 is complete, testable, crutch-free. v14.1 adds explicit definitions, fitted constants, derivation, per-galaxy methodology and falsifiable formulas addressing verification audit checklist for reproducible test.

Full simulations and wiki: [github.com/unbekanntdavid-glitch/Time-Attraction](https://github.com/unbekanntdavid-glitch/Time-Attraction)

References: Lelli+ 2016 AJ 152,157 SPARC, Bothwell+ 2022 ApJ, Planck 2020, DESI 2024, Euclid 2025 forecast. Previous Zenodo 23193655 v13, 23217189 v14.0, this v14.1 complete.

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